

02464 Artificial Intelligence and Human Cognition

Chapter 1

Human vs artificial intelligence

Cognition as computation

Human vs computer memory

Motivation

- AI models inspire models of human cognition and brain function
 - Perception and action as Bayesian inference
 - Neural encoding in visual perception
- How can we compare human and computers?
 - Human intelligence vs artificial intelligence
 - Learning
 - Memory
- Can we think of AI as a reverse engineering problem?
 - Not generally
 - Neural networks, language models and automatic speech recognition
- AI systems learn from humans
 - Mimicking human behavior
 - Image classification (typically human labels, not ground truth)
- AI systems are evaluated by humans
 - Language - who is the judge of how well GPT works?

Intelligence

- Expert definition of intelligence
 - “The ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly and learn from experience”
 - Not the same as “book learning . . . or test-taking smarts.”
 - An ability crucial for “‘catching on,’ ‘making sense’ of things, or ‘figuring out’ what to do”

Intelligence

- Many controversial issues
 - Test scores improve with instruction
 - Narrow domains of expertise
 - Book smarts versus street smarts
 - Tests are culturally biased
 - Nurture vs nature

Measuring Intelligence

- Reliability
 - Consistency
 - We expect traits like intelligence (or personality) to be relatively stable (compared to e.g. fitness or mood)
 - Measured as test-retest reliability
- Validity
 - Does the test really measure intelligence?
 - Consensus among humans decide what intelligence is
 - Correlation with math skills / income but not assembly line performance

Measuring Intelligence

- Test-retest reliability
 - Very high (around .70 or .80)—but not perfect
 - Scores can change due to
 - Measurement error
 - Environmental changes during crucial periods of development (for example, teenage years)

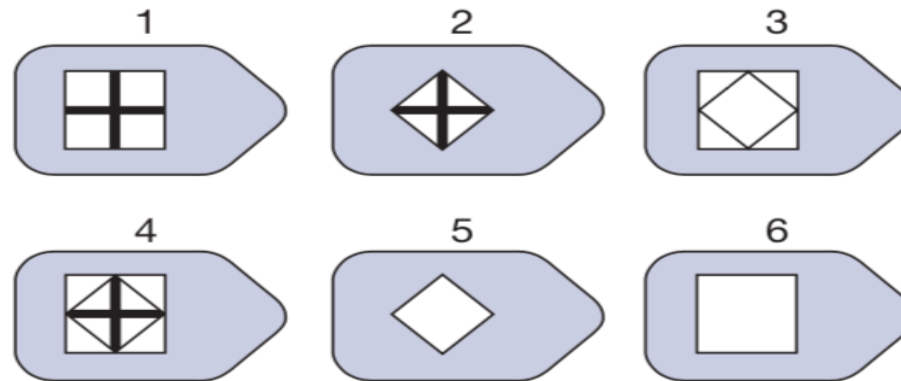
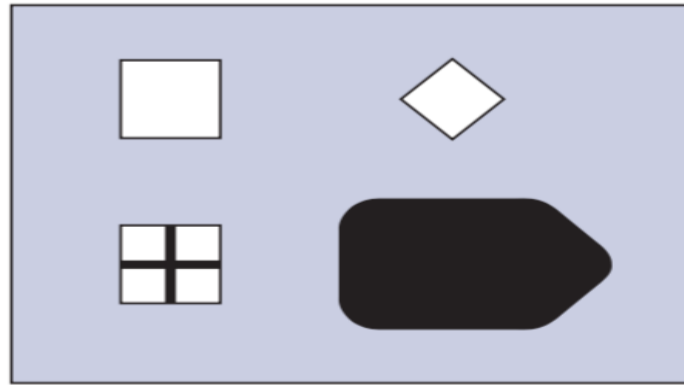
Measuring Intelligence

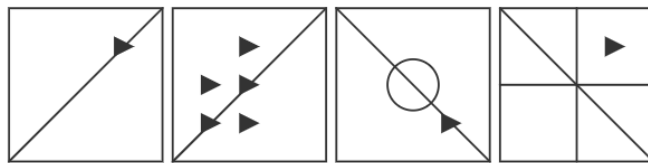
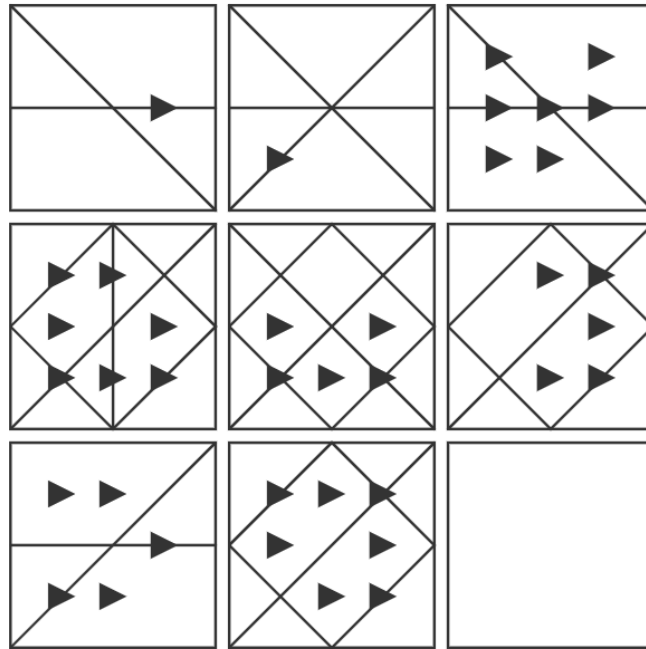
- What about validity?
 - Does the test measure what it is supposed to measure?
- Predictive validity
 - Academic performance
 - About 0.5 (means 25% of the variance)
 - Assembly line performance
 - About 0.2 (means 4% of the variance)
 - More complex professions
 - Between 0.6 and 0.8 (between 36 and 64% of the variance)

General Versus Specialized Intelligence

- General intelligence
 - One pervasive measure
- Specialized intelligence
 - Each measure is separate

Raven's progressive matrices



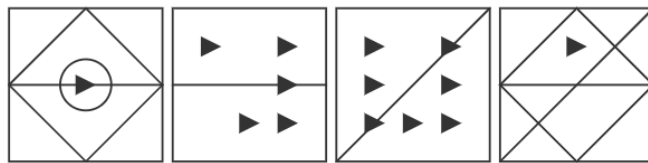


A

B

C

D



E

F

G

H

Progressive matrices

- Hard for leading AI systems in 2022
 - Less hard in 2023 <https://life architect.ai/ravens/>
- Risk of over-training
 - How many random matrices can you make?
 - How many can the AI system remember?
- Does this capture intelligence?

General Versus Specialized Intelligence



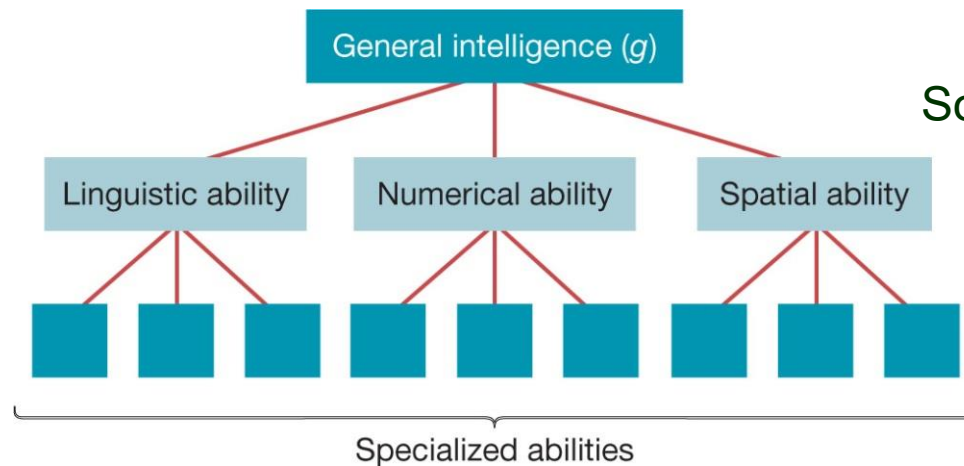
11.11 The original “Rain Man” Kim Peek (1951–2009) was a remarkable man with remarkable talents. He could read an entire book in an hour (scanning two pages at a time) and remembered everything he’d read; by one estimate, he memorized over 12,000 books. But, with an IQ of 73, he was retarded; Peek’s story was told (in fictionalized form) in the 1988 film *Rain Man*.

General Versus Specialized Intelligence

- Generalized intelligence
 - Factor analysis
 - Looks for common factors—“ingredients”
 - Reveals a general intelligence factor, or Spearman’s *g*

General Versus Specialized Intelligence

- Hierarchical intelligence



Some aspects are shared across tests

Other parts are not

Other intelligences?

- Emotional intelligence—own emotions and others' and the ability to control your emotions

Tom felt anxious and became a bit stressed when he thought about all the work he needed to do. When his supervisor brought him an additional project, he felt _____.

- a. overwhelmed
- b. depressed
- c. ashamed
- d. self-conscious
- e. jittery



Is now a good time to ask for a raise?

Other intelligences?

- Multiple intelligences (Howard Gardner)
 - Psychometric intelligence: linguistic, logical-mathematical, and spatial
 - Other types of intelligence: musical, bodily-kinesthetic, interpersonal and intrapersonal
 - May stretch the concept of *human* intelligence
 - Highly relevant for *artificial* intelligence
 - *Human* and *artificial* intelligence are *not* defined in the same way
- Definitions are important for comparing humans and computers in a meaningful way

Cognition as computing

Marr's three levels

Computational: the brain computes!

- defines the computational problem that the system is solving
 - inputs, outputs, and objective.
- It is at this level that computation in humans and machines directly are alike.
- does not specify *how* the system computes
- corresponds approximately to a set of system requirements
- Example: Estimating 3D from two 2D images

Cognition as computing

Marr's three levels

Algorithmic: *how* does the brain compute?

- defines the computer algorithms / software implementation
- Compare human and machine computation at this level
 - Computer algorithms are known by design
 - Biological algorithms are uncovered by experimentation
 - Input and output is known. Algorithm? Reverse engineering
- Example: Estimating 3D from two 2D images by triangulation

Cognition as computing

Marr's three levels

Implementational level: the physical realization of the system

- Defines the computer hardware
- Compare computers and the brain at this level
 - Computer hardware is known by design
 - Brain functions are uncovered by experimentation
 - Require measurement of brain activity
- Example: no dedicated graphics card, 4 GB RAM memory

Learning in humans and machines

- Supervised learning
 - learn a function that maps inputs (features) to outputs (labels)
 - Output can be continuous
 - Objective: minimise the error between the predicted label and the true label
 - Example: estimate 3D from 2D
 - Chapter 2: Bayesian models of perception

Learning in humans and machines

- Unsupervised learning
 - map inputs to structures
 - lower-dimensional representations (e.g. PCA/Autoencoder)
 - clusters (e.g. k-means)
 - probability distributions (e.g. GMM)
 - Objective: objective: generally, a measure of *goodness* of the structure
 - PCA/Autoencoder: compression and reconstruction
 - K-means: tightness and distinctiveness of clusters
 - GMM: likelihood of observed data
 - Example: Neural learning of representations of visual input
 - Chapter 3: Neural Encoding

Learning in humans and machines

- Reinforcement learning
 - learn a *policy*: maps states of the environment to actions
 - Objective: maximize cumulative reward across future states
 - Reward may be delayed
 - Example: learn chess from experience

Human and computer memory

- Computational level (human and computers alike)
 - the ability to encode information, store it, and then retrieve and use it
 - Guides action, decision making or processing of information
- Algorithmic level
 - Computers
 - Binary encoding of images, sound, text
 - Computers and humans
 - Associative: model parameters, weights
- Implementational level
 - Computers: binary
 - Humans: associative
 - Humans and computers: volatile and non-volatile memory

Computer volatile and non-volatile memory

- Computer volatile (working) memory (RAM)
 - Advantage: Fast access (working memory)
 - Disadvantage: Require constant energy (and expensive)
 - Low capacity
- Computer non-volatile memory (HDD, SDD)
 - Advantage: Does not require energy (and cheaper)
 - High capacity
 - Disadvantage: Slow access

Human volatile and non-volatile memory

Mini-project 1

- Algorithmic level
 - Free recall experiment
 - 20 items presented in sequence
 - Recall in any order
 - Primacy effect
 - Long-term encoding
 - Impaired by faster presentation
 - Unimpaired by secondary wm task
 - Recency effect
 - Working memory
 - Impaired by secondary wm task
 - But not by delay only
 - Unimpaired by faster presentation

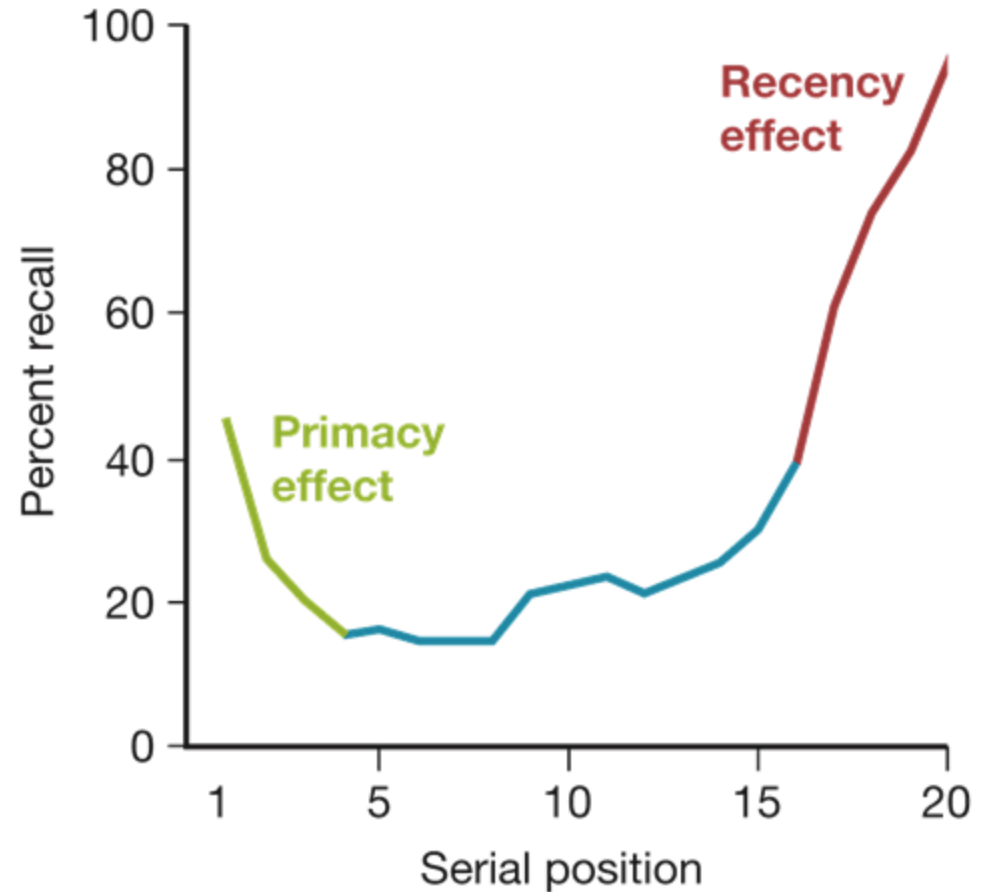


Figure 6.2
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Human volatile and non-volatile memory

- Implementational level
 - Neurons in the frontal and parietal cortex show sustained activity during working memory tasks
 - Predictive
 - Damage cause working memory problems (Patient KF)
 - Neurons in the hippocampus are active when long-term memories are stored
 - Damage cause problems with storing long-term memories (Patient HM)

Human working memory

Mini-project 1

- Algorithmic level
 - Serial recall experiment
 - Hear or see a sequence of items and report them back in the correct order
 - Capacity: 6-7 items
 - Chunking: items can be letters or three-letter words
 - Confusion errors are
 - between similar sounding letters like F and S
 - Not between similar looking letters like F and E
 - Suggests that letters are coded as sounds in memory

Human working memory

Mini-project 1

- Baddeley and Hitch's model
- Central executive
 - Encodes, decodes
- Articulatory rehearsal loop
 - Subvocalization
 - Phonological buffer (slave system)

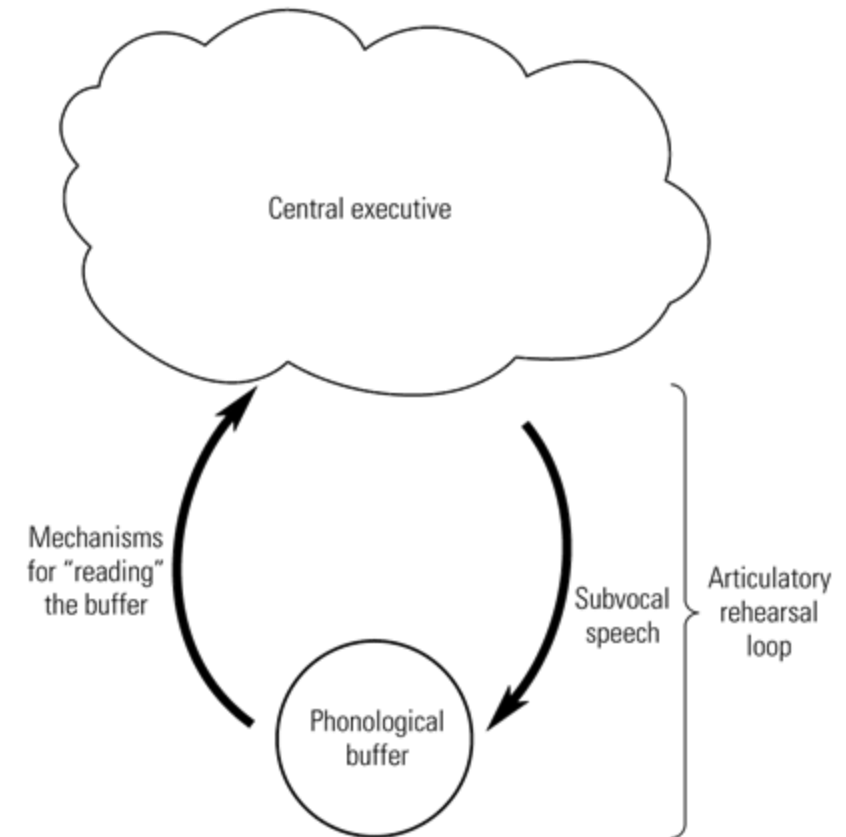


FIGURE 1.1 The Working-Memory System

Human working memory

Mini-project 1

- Baddeley and Hitch's model
- Central executive
 - Encodes, decodes
- Articulatory rehearsal loop
 - Subvocalization
 - Phonological buffer (slave system)
- Test theory
 - Articulatory suppression
 - Cut the rehearsal loop

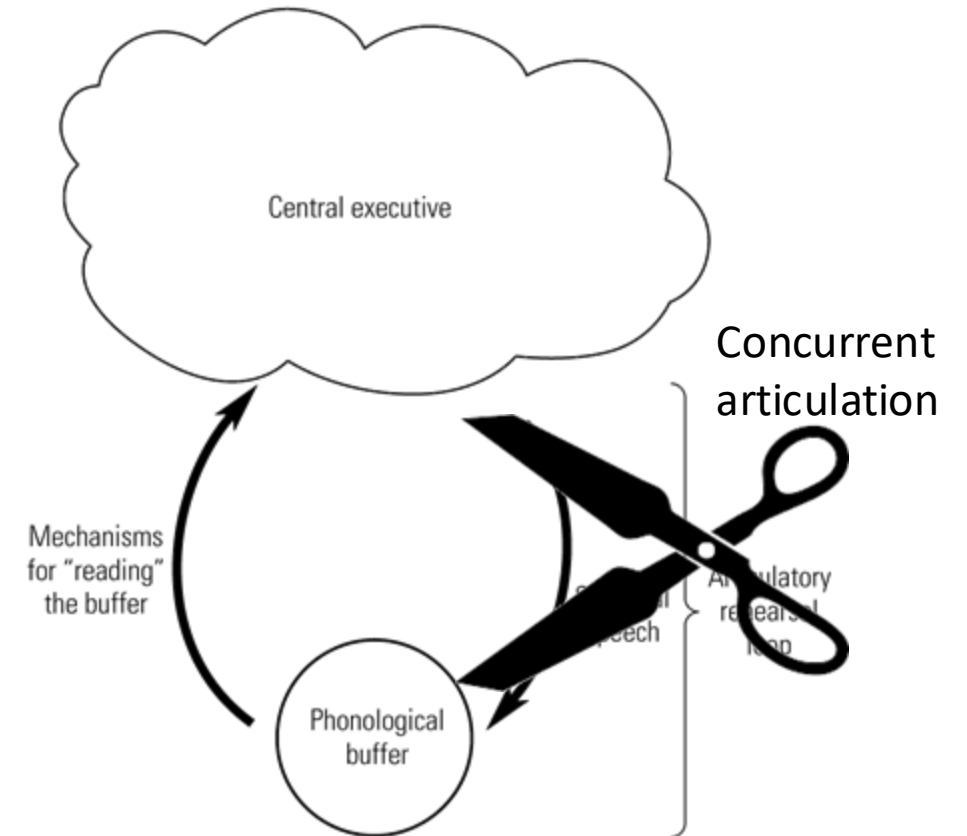


FIGURE 1.1 The Working-Memory System

Human working memory

Mini-project 1

- Testing the model experimentally
- Articulatory suppression
 - Prediction: Should disable subvocalization
 - Reduces memory span from 7 to 4 elements
 - Prediction: Should eliminate sound-alike errors
 - Result: Reduce sound-alike errors
- Specificity of Articulatory suppression
 - No effect of finger tapping
- Chunking
 - Capacity is measured in *chunks*, not letters/digits

Mini project 1 – 10% of the final grade

- Form a group of 2-4 students
- Register your group on DTU Learn
 - In DTU Learn, go to My Course -> Groups (in the course menu)
 - **23 September deadline** for registering your group
- Work on group project
 - rooms 067 and 068 in building 358
- Submit your report (max 5 pages)
 - Submission in *not* open until 24 September
 - Submission **deadline 12 October**

Mini project 1 – 10% of the final grade

- Free recall experiment effects
 - The primacy and recency effects
 - The effect of increasing the rate at which the items are presented
 - The effect of adding a working memory task to the experiment, right after the sequence ends, before the participants begin to recall it
 - The effect of adding a pause to the experiment, right after the sequence ends, before the participants begin to recall it

Mini project 1 – 10% of the final grade

- Serial recall experiment
 - The limited capacity of working memory
 - The type of errors that participants make
 - The effect of chunking
 - The effect of articulatory suppression
 - The (lack of) effect of finger tapping

Mini project 1 – 10% of the final grade

- Form a group of 2-4 students
- Register your group on DTU Learn
 - In DTU Learn, go to My Course -> Groups (in the course menu)
 - *23 September deadline*
- Read lecture notes including project description
- See you for group work in